

Experiment 7: Transfer Files from One Virtual Machine to Another

Aim

To create two lightweight Ubuntu virtual machines, connect them through a common VirtualBox network using Vagrant, verify connectivity, and transfer a file from one VM to the other.

Tools Required

- Windows host computer
- Oracle VM VirtualBox 7.x
- Vagrant 2.4.9 or compatible version
- Internet connection
- Ubuntu Vagrant box

Detailed Procedure

On Windows, verify that Vagrant is installed.

```
vagrant --version
```

Create the project directory and enter it.

```
mkdir "C:\Users\<username>\IaaS_Lab1"  
cd "C:\Users\<username>\IaaS_Lab1"
```

Initialize a Vagrant environment.

```
vagrant init
```

Create the testnet NAT network from the VirtualBox installation directory.

```
cd "C:\Program Files\Oracle\VirtualBox"  
VBoxManage natnetwork add --netname testnet --network "192.168.56.0/24" --enable --dhcp on  
VBoxManage list natnetworks
```

Return to the project directory and replace the generated Vagrantfile with the following configuration.

```
Vagrant.configure("2") do |config|  
  config.vm.define "vm1" do |vm1|  
    vm1.vm.box = "ubuntu/bionic64"  
    vm1.vm.hostname = "vm1"  
    vm1.vm.network "public_network", bridge: "testnet", ip: "192.168.56.10"  
    vm1.vm.provider "virtualbox" do |vb|  
      vb.memory = "1024"; vb.cpus = 1; vb.name = "vm1"  
    end  
  end  
  
  config.vm.define "vm2" do |vm2|  
    vm2.vm.box = "ubuntu/bionic64"  
    vm2.vm.hostname = "vm2"  
    vm2.vm.network "public_network", bridge: "testnet", ip: "192.168.56.11"  
    vm2.vm.provider "virtualbox" do |vb|  
      vb.memory = "1024"; vb.cpus = 1; vb.name = "vm2"  
    end  
  end  
end
```

Start both VMs.

```
vagrant up
```

Verify that both machines are running.

```
vagrant status
```

Enter VM1 and verify its network address.

```
vagrant ssh vm1  
ip addr show
```

Confirm that the second interface has 192.168.56.10/24. Exit VM1.

```
exit
```

Enter VM2 and verify its network address.

```
vagrant ssh vm2  
ip addr show
```

Confirm that the second interface has 192.168.56.11/24.

From VM2, test connectivity to VM1.

```
ping 192.168.56.10
```

After confirming replies, press Ctrl + C to stop the ping.

For the actual file-transfer demonstration, create a file on VM1.

```
echo "Hello from VM1 - Cloud Computing Lab 7" > transfer.txt  
cat transfer.txt
```

Keep VM1 available and start a temporary HTTP server in VM1.

```
python3 -m http.server 8000
```

Open a second Windows Command Prompt, enter VM2, and download the file from VM1.

```
cd "C:\Users\<username>\IaaS_Lab1"  
vagrant ssh vm2  
wget http://192.168.56.10:8000/transfer.txt
```

Verify the transferred file on VM2.

```
cat transfer.txt
```

Important Compatibility / Networking Note

The first vagrant up displayed a warning that the named bridge testnet was not found, but the resulting VMs still received the required 192.168.56.10 and 192.168.56.11 addresses on their second interfaces. The final connectivity test succeeded with 0% packet loss. The procedure records the working configuration and verifies the actual IP addresses before testing transfer.

Output

```
vagrant@ubuntu:~$ ssh vm1
C:\Users\Afrah M\IaaS_Lab1>vagrant ssh vm1
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-212-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Mon Sep 14 09:25:46 UTC 2026

System load:  0.14          Processes:      89
Usage of /:   2.9% of 38.7GB Users logged in:    0
Memory usage: 12%          IP address for enp0s3: 10.0.2.15
Swap usage:   0%           IP address for enp0s8: 192.168.56.10

Expanded Security Maintenance for Infrastructure is not enabled.
0 updates can be applied immediately.

Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '20.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

vagrant@vm1:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 02:3b:7b:b7:3b:2d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86135sec preferred_lft 86135sec
    inet6 fd17:625c:f037:2:3b:7bff:feb7:3b2d/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86348sec preferred_lft 14348sec
    inet6 fe80::3b:7bff:feb7:3b2d/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:bd:e5:ac brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.10/24 brd 192.168.56.255 scope global enp0s8
        valid_lft forever preferred_lft forever
    inet6 2406:7400:ce:82ec:a00:27ff:feb7:3b2d/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 6936sec preferred_lft 297sec
    inet6 fe80::a00:27ff:feb7:3b2d/64 scope link
        valid_lft forever preferred_lft forever
vagrant@vm1:~$ exit
logout

vagrant@ubuntu:~$ ssh vm2
C:\Users\Afrah M\IaaS_Lab1>vagrant ssh vm2
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-212-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Mon Sep 14 09:27:28 UTC 2026

System load:  0.15          Processes:      88
Usage of /:   2.9% of 38.7GB Users logged in:    0
Memory usage: 12%          IP address for enp0s3: 10.0.2.15
Swap usage:   0%           IP address for enp0s8: 192.168.56.11

Expanded Security Maintenance for Infrastructure is not enabled.
0 updates can be applied immediately.

Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '20.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

vagrant@vm2:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 02:3b:7b:b7:3b:2d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86189sec preferred_lft 86189sec
    inet6 fd17:625c:f037:2:3b:7bff:feb7:3b2d/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86189sec preferred_lft 14189sec
    inet6 fe80::3b:7bff:feb7:3b2d/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:72:31:fb brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.11/24 brd 192.168.56.255 scope global enp0s8
        valid_lft forever preferred_lft forever
    inet6 2406:7400:ce:82ec:a00:27ff:fe72:31fb/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 6990sec preferred_lft 281sec
    inet6 fe80::a00:27ff:fe72:31fb/64 scope link
        valid_lft forever preferred_lft forever
vagrant@vm2:~$ ping 192.168.56.10
PING 192.168.56.10 (192.168.56.10) 56(84) bytes of data.
```

```
vagrant@vm1:~$ ping 192.168.56.10
PING 192.168.56.10 (192.168.56.10) 56(84) bytes of data.
64 bytes from 192.168.56.10: icmp_seq=1 ttl=64 time=2.89 ms
64 bytes from 192.168.56.10: icmp_seq=2 ttl=64 time=1.40 ms
64 bytes from 192.168.56.10: icmp_seq=3 ttl=64 time=1.10 ms
64 bytes from 192.168.56.10: icmp_seq=4 ttl=64 time=3.40 ms
64 bytes from 192.168.56.10: icmp_seq=5 ttl=64 time=1.18 ms
64 bytes from 192.168.56.10: icmp_seq=6 ttl=64 time=1.28 ms
64 bytes from 192.168.56.10: icmp_seq=7 ttl=64 time=0.913 ms
64 bytes from 192.168.56.10: icmp_seq=8 ttl=64 time=1.40 ms
64 bytes from 192.168.56.10: icmp_seq=9 ttl=64 time=1.76 ms
--- 192.168.56.10 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8034ms
rtt min/avg/max/mdev = 0.913/1.729/3.400/0.820 ms
vagrant@vm1:~$ exit
Logout
C:\Users\Afrah M\IaaS_Lab1>vagrant ssh vm1
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-212-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

System information as of Mon Sep 14 09:29:12 UTC 2026

System load:  0.17               Processes:    86
Usage of /:   2.9% of 38.70GB    Users logged in: 0
Memory usage: 12%               IP address for enp0s3: 10.0.2.15
Swap usage:   0%                IP address for enp0s8: 192.168.56.10

Expanded Security Maintenance for Infrastructure is not enabled.

0 updates can be applied immediately.

Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '20.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Mon Sep 14 09:25:47 2026 from 10.0.2.2
vagrant@vm1:~$ echo "Hello from VM1 - Cloud Computing Lab 7" > transfer.txt
vagrant@vm1:~$ cat transfer.txt
Hello from VM1 - Cloud Computing Lab 7
vagrant@vm1:~$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
192.168.56.11 - - [14/Sep/2026 09:30:52] "GET /transfer.txt HTTP/1.1" 200 -
C
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Afrah M>cd "C:\Users\Afrah M\IaaS_Lab1"

C:\Users\Afrah M\IaaS_Lab1>vagrant ssh vm2
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-212-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

System information as of Mon Sep 14 09:30:35 UTC 2026

System load:  0.0               Processes:    86
Usage of /:   2.9% of 38.70GB    Users logged in: 0
Memory usage: 12%               IP address for enp0s3: 10.0.2.15
Swap usage:   0%                IP address for enp0s8: 192.168.56.11

Expanded Security Maintenance for Infrastructure is not enabled.

0 updates can be applied immediately.

Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '20.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Mon Sep 14 09:27:29 2026 from 10.0.2.2
vagrant@vm2:~$ wget http://192.168.56.10:8000/transfer.txt
--2026-09-14 09:30:51--  http://192.168.56.10:8000/transfer.txt
Connecting to 192.168.56.10:8000... connected.
HTTP request sent, awaiting response... 200 OK
Length: 39 [text/plain]
Saving to: 'transfer.txt'

transfer.txt          100%[=====] 39 --.-KB/s  in 0s

2026-09-14 09:30:51 (3.85 MB/s) - 'transfer.txt' saved [39/39]

vagrant@vm2:~$ cat transfer.txt
Hello from VM1 - Cloud Computing Lab 7
vagrant@vm2:~$ exit
Logout
C:\Users\Afrah M\IaaS_Lab1>
```

Result

Two lightweight Ubuntu VMs were created and connected to the same 192.168.56.0/24 network. VM2 successfully communicated with VM1 and successfully retrieved the file transfer.txt from VM1.